Mathematical Analysis 2

Multivariable derivatives

Peter Zeman

Throughout, vectors are columns and all norms are Euclidean. Problems marked $\star$ are further practice.

The derivative is a linear map

Let $U \subseteq \mathbb{R}^n$ be open, $F : U \to \mathbb{R}^m$, and $a \in U$. We say that F is **differentiable at** a if there is a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{h \to 0} \frac{\|F(a+h) - F(a) - L(h)\|}{\|h\|} = 0.$$

We write $DF(a) = L$. The derivative is a map acting on an increment h; $F(a)$ is a value.

1 Constant, linear, and affine maps

Use the definition above, rather than coordinate differentiation.

(a) Find $DF(a)h$ and the remainder $F(a+h) - F(a) - DF(a)h$ for

$$F(x) = c, \qquad F(x) = Ax, \qquad F(x) = Ax + b,$$

where $A \in \mathbb{R}^{m \times n}$ and $b, c \in \mathbb{R}^m$ are fixed. Give the matrix of $DF(a)$ in the standard bases and its size. Does it depend on a?

(b) Take

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \quad a = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad h = \begin{pmatrix} 0.02 \\ -0.01 \\ 0.03 \end{pmatrix}.$$

Compute $F(a)$, $DF(a)h$, and $F(a+h)$. Is the linear approximation exact?

(c) $\star$ Prove that the linear map in the definition is unique. *Hint: compare two candidates on increments $h = tv$.*

2 Quadratic forms from the definition

Let $B \in \mathbb{R}^{n \times n}$ be fixed and $Q(x) = x^{\mathsf{T}} B x$.

(a) Expand $Q(a+h) - Q(a)$. Identify the part linear in h and prove that the remaining term, divided by $\|h\|$, tends to zero. Give $DQ(a)h$ and the matrix of $DQ(a)$ in the standard bases, without assuming that B is symmetric.

(b) Simplify your answer when $B = B^{\mathsf{T}}$. Apply it to

$$Q(x, y) = 2x^2 - 2xy + 3y^2, \qquad a = (1, -1).$$

Use the derivative to approximate $Q(1.02, -0.99)$. Compute the exact remainder and compare with the exact value.

Partial derivatives and directions

Derivatives along a line

For $F : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ and $v \in \mathbb{R}^n \setminus \{0\}$, define

$$D_v F(a) = \lim_{t \to 0} \frac{F(a + tv) - F(a)}{t}, \qquad \frac{\partial F}{\partial x_i}(a) = D_{e_i} F(a),$$

whenever the limits exist. Limits are **two-sided**. We do **not** automatically normalize v. For a scalar-valued function, the **gradient** is the column vector $\nabla f(a) = (\frac{\partial f}{\partial x_1}(a), \ldots, \frac{\partial f}{\partial x_n}(a))^{\mathsf{T}}$.

3 From the total derivative to partial derivatives

Suppose that $F : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at a.

(a) Prove directly from the definition that

$$D_v F(a) = DF(a)v.$$

(b) For $F = (f_1, \ldots, f_m)^{\mathsf{T}}$, the **Jacobian matrix** $J_F(a)$ is the $m \times n$ matrix with (j, i) entry $\frac{\partial f_j}{\partial x_i}(a)$. Prove that it represents $DF(a)$ in the standard bases. Deduce, for scalar-valued f, that $D_v f(a) = \nabla f(a)^{\mathsf{T}} v$.

(c) Find $\nabla Q(1, -1)$ for the quadratic form in Problem 2, and its directional derivative in the unit direction $v = (3/5, 4/5)$.

4 Computing partial derivatives

For each function, find its open domain, compute all first partial derivatives, and evaluate the gradient at the indicated point. Explain why the function is differentiable there. Then compute $D_v f(a)$.

(a) $f(x, y) = e^{xy}, \quad a = (1, 0), \quad v = (1, -1).$

(b) $f(x, y) = \ln(x^2 - 2x - y), \quad a = (0, -1), \quad v = (1, 2).$

(c) $f(x, y, z) = (x/y)^z, \quad a = (e, 1, 2), \quad v = (0, 1, -1).$ Interpret a variable real power by $u^z = \exp(z \ln u)$ for $u > 0$.

5 Why the gradient gives steepest ascent and descent

Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ be differentiable at a.

(a) Assume $\nabla f(a) \neq 0$. Prove that among *unit* vectors, the unique direction of steepest ascent is $\nabla f(a) / \|\nabla f(a)\|$, and the unique direction of steepest descent is its negative. Find the corresponding rates of change. *Hint: use Cauchy–Schwarz and its equality condition.*

(b) What happens if $\nabla f(a) = 0$? Does this imply that a is a local maximum or minimum? Give a counterexample if not.

(c) Apply the result to $g(x, y) = x^2 + 4y^2$ at $(1, 1)$. Also compute the derivatives for $v = (1, 0), (0, 1), (1, 1)$ and $(1, 1)/\sqrt{2}$. Why must directions have equal length when comparing rates of change per unit distance?

Norms, composition, and linear approximation

6 The squared norm, the norm, and its reciprocal

Consider $S, N : \mathbb{R}^n \to \mathbb{R}$, where $n \geq 1$, given by

$$S(x) = \|x\|^2, \qquad N(x) = \|x\|.$$

(a) Use the definition to find $DS(a)h$ at every $a \in \mathbb{R}^n$.

(b) For $a \neq 0$, use the definition to prove that

$$DN(a)h = \frac{\langle a, h\rangle}{\|a\|}.$$

Give the gradient and bound the remainder by a constant times $\|h\|^2$ near $h = 0$. *Hint: rationalize $\|a+h\| - \|a\|$ and use the reverse triangle inequality.*

(c) Is N differentiable at 0? Does $D_v N(0)$ exist for a nonzero v under our two-sided convention? Compare with S.

(d) Find the derivative and gradient of $R(x) = 1/\|x\|$ on $\mathbb{R}^n \setminus \{0\}$.

7 Using the chain rule

You may use the chain rule

$$D(G \circ F)(a) = DG(F(a)) \circ DF(a), \qquad J_{G\circ F}(a) = J_G(F(a))J_F(a).$$

State the input and output dimensions of the maps you compose.

(a) **A first composition.** Let $F : \mathbb{R}^2 \to \mathbb{R}^2$ and $G : \mathbb{R}^2 \to \mathbb{R}$ be given by

$$F(x, y) = \begin{pmatrix} x + y \\ xy \end{pmatrix}, \qquad G(u, v) = u^2 + v.$$

Write down $H = G \circ F$. Compute $J_F(x, y)$ and $J_G(u, v)$, and use their product to find $J_H(x, y)$ and $J_H(1, 2)$. Check your answer by differentiating the explicit formula for H.

(b) **A tangent plane.** Let $f(x, y) = \sqrt{x^2 + 4y^2 + 1}$. Find $Df(1, 1)$ and the tangent plane to its graph at $(1, 1, f(1, 1))$. Use it to approximate $f(1.02, 0.99)$.

(c) **Least squares.** Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$. For

$$L(x) = \tfrac{1}{2}\|Ax - b\|^2,$$

find $DL(a)h$ and $\nabla L(a)$. Write the equation $\nabla L(a) = 0$ in matrix form.

(d) ⋆ **Normalizing a vector.** Let $T : \mathbb{R}^n \setminus \{0\} \to \mathbb{R}^n$, $T(x) = x/\|x\|$. Find $DT(a)h$ and its Jacobian. Show that the radial direction a lies in the kernel and that $DT(a)h$ is perpendicular to a.

Applications and further practice

What the approximation says

For a differentiable map,

$$F(a+h) = F(a) + DF(a)h + r(h), \qquad \frac{\|r(h)\|}{\|h\|} \longrightarrow 0.$$

The linear term predicts small changes. A first-order error estimate is not automatically a rigorous bound for a fixed, nonzero measurement error.

8 ⋆ Measurement errors

Use the *measured* values as nominal values. The stated error bounds may be attained simultaneously. Use radians for angles.

(a) **A cylinder.** The volume is $V(r,h) = \pi r^2 h$. The measured radius is $r_0 = 3\,\text{cm}$, with $|\Delta r| \le 0.01\,\text{cm}$, and the measured height is $h_0 = 5\,\text{cm}$, with $|\Delta h| \le 0.005\,\text{cm}$. Use the differential to estimate the largest absolute and relative error in the computed volume. Then find the exact largest absolute error by comparing the endpoint volumes.

(b) **A triangle.** Its area is

$$P(b,c,\alpha) = \tfrac{1}{2} bc \sin\alpha.$$

The measured values $b_0, c_0 > 0$ and $\alpha_0 = \pi/4$ each have relative error at most 1%, for example $|\Delta\alpha| \le 0.01\alpha_0$. Let $P_0 = P(b_0, c_0, \alpha_0)$. Derive an expression for dP/P_0 and show that the *first-order* estimate of the largest relative error is less than 2.8%.

(c) Does part (b) give a strict 2.8% bound on the *exact* relative error $|P - P_0|/P_0$? Find the exact largest relative error over the allowed intervals. *Hint: the area increases with each variable on these intervals.*

9 ⋆ What partial and directional derivatives do not tell us

(a) Define

$$H(x,y) = \begin{cases} \dfrac{x^3}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Prove that H is continuous at the origin and compute every $D_v H(0,0)$ for $v = (p,q) \neq 0$. Find both partial derivatives at the origin. Is H differentiable there? Explain why computing $\nabla H(0,0)^{\mathsf{T}} v$ can give the wrong answer here.

(b) Define

$$K(x,y) = \begin{cases} (x^2+y^2)\sin\!\left(\dfrac{1}{x^2+y^2}\right), & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Prove from the definition that K is differentiable at the origin. Compute its partial derivatives away from the origin and show that they are not continuous at the origin. What does this say about the condition that the first partial derivatives be continuous near a point?